CHRIST (Deemed to be University), Bangalore
School of Engineering and Technology
Department of Computer Science and Engineering

COURSE PLAN

Odd Semester/2026-27

Semester	III	Programme & Class	M. Tech / 3MTCS
Course Code	MTCS344E01	Course Type	PEC
Course Title	Advanced Data Mining	LCA	No
Total Hours	45	Hours per week	3
Faculty Name	Dr Manohar M	Contact Details and Office Hours	Cabin No. 07, Ground Floor, 2nd Block, SoET, Kengeri Campus Email: manohar.m@christuniversity.in
Class Policies and Guidelines	• Mobile phones and Laptops may be used as per the instructions given by the faculty. • Students are requested to adhere on the CIA deadline and assignment submission. • Failing which the re test or re-submission can be considered only for 80% of marks • Mode of communication to students through university mail and LMS only.		
Course Description	This course provides an understanding of basic fundamentals, methodologies and applications of data mining. It explores both supervised and unsupervised learning methods. Through practical applications, students will gain the knowledge to solve real world problems.		
Course Objectives (Optional)	1. To gain a comprehensive understanding of data mining fundamentals and develop expertise in data preprocessing techniques. 2. To explore and apply the supervised and unsupervised learning methods in data mining.		

	3. To design and implement models for solving complex real-world problems effectively.			
Course Outcomes	**CO1:** Explain the basic fundamentals of Data mining and how to preprocess the data. (L2) **CO2:** Apply appropriate association algorithm to mine frequent patterns.(L3) **CO3:** Apply classification and related data mining techniques accurately and efficiently using Python for computations.(L3) **CO4:** Apply clustering and related data mining techniques accurately and efficiently using Python for computations. (L3) **CO5:** Develop a model using appropriate algorithm to solve real world problems.(L3)			
SECTION II				
Week	Module/ Unit/ Topic details	Andragogy	Ungraded formative Assessments	Resource/ Reference details
1	**Unit 1 : Introduction to Data Mining:** Data Mining: Introduction to Data Mining- Steps in Data Mining Process-Major Issues in Data Mining- Working with Data: Scraping the Web	NA	Question and answer	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis: Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python
2	**Unit 1 : Introduction to Data Mining:** Data Preprocessing: An Overview- Data Cleaning-	NA	Discussion	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis:

	Data Integration- Data Reduction			Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python
3	**Unit 1 : Introduction to Data Mining:** Data Transformation. **Practical Experiments / Experiential Learning:** Ex 1:Write a python program to scrape the details of most downloaded articles from AI in last 90 days https://www.journals.elsevier.com/artificial-intelligence/most-downloaded-articles Scrape below mentioned details: Paper Title	Experiential Learning	Question and answer	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis: Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python

	Authors Published Date Paper URL Ex 2: Write a Python program to perform data cleaning on a sample dataset. The program should handle tasks such as handling missing values, removing duplicate entries and dealing with outliers.			
4	**Unit 2 :Frequent Pattern Mining:** Mining Frequent Patterns-Basic Concepts-Efficient and scalable frequent item set mining methods: Apriori Algorithm	NA	Quiz	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis: Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python
5	**Unit 2 :Frequent Pattern Mining:** FP-Growth Algorithm-Advanced	NA	Question and answer	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis:

	Pattern Mining: Pattern Mining in Multilevel, Multidimensional Space-Constraint-Based Frequent Pattern Mining			Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python
6	**Unit 2 :Frequent Pattern Mining:** Mining HighDimensional Data and Colossal Patterns. **Practical Experiments / Experiential Learning:** Ex 1:Write a python program to implement the apriori algorithm for frequent itemset mining. Ex 2: Write a python program to mine frequent patterns using the FP-Growth algorithm.	Experiential Learning	Concept Mapping	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis: Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python

7	**Unit 3 : Classification:** Basic Concepts-Classification by Decision Tree Induction-Advanced classification methods:Bayesian Belief Networks-Classification by Backpropagation	NA	Quiz	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis: Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python
8	**Unit 3 : Classification:** Support Vector Machines-k NearestNeighbourClassifiers-Genetic Algorithms-Rough Set Approach-Fuzzy Set-Model Evaluation and SelectionApproaches	NA	Question and answer	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis: Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python
9	**Unit 3 : Classification:** Techniques to Improve Classification Accuracy. **Practical Experiments /**	Experiential Learning	Concept Mapping	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis: Fundamental concepts and algorithms

	Experiential Learning: Ex 1: Write a python program to classify data using SVM Ex 2:Write a python program to implement a Multilayer Perceptron for classification. Ex 3: Write a python program to implement K-Nearest Neighbor Classification Algorithm			R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python
10	**Unit 4 : Cluster Analysis:** Basic Concepts and Methods- Partitioning Methods: k-Means- k-Medoids	NA	Quiz	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis: Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python

11	**Unit 4 : Cluster Analysis:** Hierarchical Methods- Advanced Cluster Analysis: Probabilistic Model Based Clustering, Clustering HighDimensional Data, Clustering Graph and Network Data.	NA	Question and answer	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis: Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python
12	**Unit 4 : Cluster Analysis:** Evaluation of Clustering. **Practical Experiments / Experiential Learning:** Ex 1:Write a python program to implement K-Means Clustering Algorithm for a Sample dataset. (K=3) Ex 2: Write a python program to implement K-Medoid Clustering Algorithm for	Experiential Learning	Concept Mapping	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis: Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python

	a Sample dataset (K=3).			
13	**Unit 5 : Outlier Detection and Data Mining Applications:** Outliers and Outlier Analysis- Outlier Detection Methods- ProximityBased Approaches- ClusteringBased Approaches	NA	Quiz	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis: Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python
14	**Unit 5 : Outlier Detection and Data Mining Applications:** Detection in High Dimensional Data- Data Mining Applications: Recommender Systems Intrusion Detection and Prevention	NA	Question and answer	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis: Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python
15	**Unit 5 : Outlier Detection and Data Mining Applications:**	Experiential Learning	Concept Mapping	T1: Data Mining: Concepts and Techniques. R1: Introduction to Data Mining R2: Data Mining and Analysis:

	Outlier Financial Data Analysis. **Practical Experiments / Experiential Learning:** Ex 1: Develop a mini project to apply data mining techniques using python to solve real world problems.			Fundamental concepts and algorithms R3: Data Mining – Practical Machine Learning Tools and Techniques R4: Data Science from Scratch: First Principles with Python

SECTION I

SECTION III

Note: Refer to the syllabus entry in Espro (Except for CO-PO strength mapping, rest of the details will be populated based on your Espro Syllabus entry)

Mapping:

Programme Outcomes	CO1	CO2	CO3	CO4	CO5
PO1	1	1	3	3	3
PO2	1	2	3	3	3
PO3	-	-	3	3	3
PO4	-	-	-	3	3
PO5	-	-	-	-	3
PO6	-	-	-	-	-
PO7	-	-	-	-	-
PO8	-	-	-	-	-
PO9	-	-	-	-	-
PO10	-	-	-	-	-
PO11	-	-	-	-	-
PO12	-	-	-	-	-

NOTE: Please add /delete rows and columns where necessary to accommodate all the Programme Outcomes and Programme Specific Outcomes.

Mapping of needs with the syllabus:

Course Outcomes	Local, Regional, National, Global and Development Needs (LRNG)	Cross-cutting issues relating to Professional Ethics, Gender, Human Values, Environment and Sustainability	Employability/Entrepreneurship/ Skill Development	Sustainable Development Goals
CO1	Local		Skill Development	
CO2	Regional		Employability	
CO3	National		Employability	
CO4	Global		Employability	
CO5	National		Skill Development	
Topic/Module Details				
I	Scraping the Web, Data Preprocessing: An Overview- Data Cleaning- Data Integration- Data Reduction - Data Transformation	-	Data Mining: Introduction to Data Mining- Steps in Data Mining Process- Major Issues in Data Mining- Working with Data	-

II	Mining Frequent Patterns-Basic Concepts-Efficient and scalable frequent item set mining methods: Multidimensional Space-Constraint-Based Frequent Pattern Mining-Mining High Dimensional Data and Colossal Patterns.	-	Apriori Algorithm-FP-Growth Algorithm-Advanced Pattern Mining: Pattern Mining in Multilevel	-
III	Basic Concepts-Classification, Rough Set Approach-Fuzzy Set-Model Evaluation and SelectionApproaches-Techniques to Improve Classification Accuracy.	-	Decision Tree Induction-Advanced classification methods:Bayesian Belief Networks-Classification by Backpropagation- Support Vector Machines-k NearestNeighbourClassifiers-Genetic Algorithms	-

IV	Basic Concepts and Methods- Partitioning Methods Evaluation of Clustering.	-	: k-Means- k-Medoids, Hierarchical Methods- Advanced Cluster Analysis: Probabilistic Model Based Clustering, Clustering HighDimensional Data, Clustering Graph and Network Data.	-
V	Outliers and Outlier Analysis- Outlier Detection Methods	-	Proximity Based Approaches- Clustering Based Approaches- Outlier Detection in High Dimensional Data- Data Mining Applications: Recommender Systems- Intrusion Detection and Prevention- Financial Data Analysis.	-

A template to map the Course Outcomes against the components of the assessment is given below:

Course Outcomes	Components of assessment (with the breakup of marks)					
	CAC-1- (10 marks)	CAC-2 (10 marks)	TEST-1 (50 marks)	CAC-3 (10 marks)	CAC-4 (10 marks)	TEST-2 (50 marks)
CO 1	10	NA	20	NA	NA	NA
CO 2	NA	10	20	NA	NA	NA
CO 3	NA	NA	10	NA	NA	10
CO 4	NA	NA	NA	10	NA	20
CO 5	NA	NA	NA	NA	10	20

Note: Add more rows if there are more COs

Assessment outline:

	Components of assessment This will be populated from the syllabus					
Components	CAC-1	CAC-2	TEST-1	CAC-3	CAC-4	TEST-2
Marks/Percentage	12	13	25	12	13	25

SECTION IV

CAC-1

Assignment Component: Closed Book Test

Assignment Description:

To evaluate the student's knowledge on:

a. Comprehension of technical aspects of the Data Mining domain.
b. Demonstrate the capability to apply basics of Data Mining.

Evaluation Criteria/Rubric/s:

Criteria					
Area of Evaluation	Excellent (9 to10)	Very Good (7 to 8)	Good (5 to 6)	Satisfactory (3 to 4)	Needs Improvement (2 to 0)
Attempt made for the question(2.5 marks)	If the student has attempted the entire question including the sub-categories in the question he/she is eligible to get the marks of this range.	If the student has attempted the partial question including the sub-categories in the question he/she is eligible to get the marks of this range.	If the student has attempted the exact half of the question including the sub-categories in the question he/she is eligible to get the marks of this range.	If the student has attempted only a part if the given question including the sub-categories in the question he/she is eligible to get the marks of this range.	If the student has not attempted the entire/ any part question including the sub-categories in the question he/she is eligible to get the marks of this range.
Correctness of the answer and relevance(2.5 marks)	If the question attempted by the students is entirely correct when considered all the sub-categories in the question he/she is eligible to get the marks of this range.	If the question attempted by the students is partially correct when considered all the sub-categories in the question he/she is eligible to get the marks of this range.	If the question attempted by the students is exact half the way correct when considered all the sub-categories in the question he/she is eligible to get the marks of this range.	If the question attempted by the students is correct only in a small part when considered all the sub-categories in the question he/she is eligible to get the marks of this range.	If the question attempted by the students is not correct when considered all the sub-categories in the question he/she is eligible to get the marks of this range.
Approach followed to reach the solution(2.5 marks)	If the student has followed the expected and correct approach to attempt the question right he/she is eligible to get the marks of this range.	If the student has followed the partial approach to attempt the question right he/she is eligible to get the marks of this range.	If the student has followed the half erroneous approach to attempt the question right he/she is eligible to the marks of this range.	If the student has followed the improper (partly) approach to attempt the question right he/she is eligible to get the marks of this range.	If the student has followed the wrong approach to attempt the question right he/she is eligible to get the marks of this range.

Legibility(2.5 marks)	If the student has attempted the question with complete correctness on legible writing he/she is eligible to get the marks of this range.	If the student has attempted the question with complete correctness on satisfactory writing he/she is eligible to get the marks of this range.	If the student has attempted the question with complete correctness on not satisfactory writing he/she is eligible to get the marks of this range.	If the student has attempted the question with complete correctness on illegible writing he/she is eligible to get the marks of this range.	If the student has attempted the question with complete correctness on poor writing he/she is eligible to get the marks of this range.

CAC-2

Assignment Component: Problem Solving Assignment

Assignment Description:

To evaluate the student's knowledge on:

a. Frequent pattern mining

Evaluation Criteria/Rubric/s:

Criteria	**Excellent** 9 to 10	**Good** 7 to 8	**Fair** 5 to 6	**Needs Improvement** 0 to 4
Analysis and understanding(2 marks)	Demonstrates a thorough analysis of the problem; considers all relevant factors and underlying causes.	Provides a good analysis of the problem; considers most relevant factors and underlying causes.	Provides a basic analysis of the problem; considers some relevant factors but misses others.	Provides a superficial or incorrect analysis of the problem; overlooks important factors and underlying causes.
Strategy Development(2 marks)	Develops a clear, logical, and innovative strategy to solve the problem; considers multiple approaches.	Develops a clear and logical strategy; considers alternative approaches but with minor gaps.	Develops a basic strategy to solve the problem; considers limited alternative approaches.	Develops an unclear or illogical strategy; fails to consider alternative approaches.

Creativity and Innovation(2 marks)	Demonstrates exceptional creativity and innovation in solving the problem; thinks outside the box.	Demonstrates good creativity and innovation; some original ideas and approaches.	Demonstrates some creativity and innovation; relies mostly on conventional ideas and approaches.	Lacks creativity and innovation; relies solely on conventional ideas and approaches.
Critical Thinking(2 marks)	Applies critical thinking consistently; evaluates all possible solutions and anticipates potential issues.	Applies critical thinking frequently; evaluates most solutions and anticipates some potential issues.	Applies critical thinking occasionally; evaluates some solutions but misses others and potential issues.	Lacks critical thinking; fails to evaluate solutions thoroughly or anticipate potential issues.
Evaluation(2 marks)	Thoroughly evaluates the effectiveness of the solution.	Evaluates the effectiveness of the solution with minor gaps.	Provides a basic evaluation of the solution's effectiveness.	Fails to evaluate the solution's effectiveness.

CAC-3

Assignment Component: Closed Book Test

Assignment Description:

To evaluate the student's knowledge on:

a. Comprehension of technical aspects of the Clustering Techniques.
b. Demonstrate the capability to apply basics of Clustering techniques techniques.

Evaluation Criteria/Rubric/s:

Criteria					
Area of Evaluation	Excellent (9 to 10)	Very Good (7 to 8)	Good (5 to 6)	Satisfactory (3 to 4)	Needs Improvement (0 to 2)

Attempt made for the question(2.5 marks)	If the student has attempted the entire question including the sub-categories in the question he/she is eligible to get the marks of this range.	If the student has attempted the partial question including the sub-categories in the question he/she is eligible to get the marks of this range.	If the student has attempted the exact half of the question including the sub-categories in the question he/she is eligible to get the marks of this range.	If the student has attempted only a part if the given question including the sub-categories in the question he/she is eligible to get the marks of this range.	If the student has not attempted the entire/ any part question including the sub-categories in the question he/she is eligible to get the marks of this range.
Correctness of the answer and relevance(2.5 marks)	If the question attempted by the students is entirely correct when considered all the sub-categories in the question he/she is eligible to get the marks of this range.	If the question attempted by the students is partially correct when considered all the sub-categories in the question he/she is eligible to get the marks of this range.	If the question attempted by the students is exact half the way correct when considered all the sub-categories in the question he/she is eligible to get the marks of this range.	If the question attempted by the students is correct only in a small part when considered all the sub-categories in the question he/she is eligible to get the marks of this range.	If the question attempted by the students is not correct when considered all the sub-categories in the question he/she is eligible to get the marks of this range.
Approach followed to reach the solution(2.5 marks)	If the student has followed the expected and correct approach to attempt the question right he/she is eligible to get the marks of this range.	If the student has followed the partial approach to attempt the question right he/she is eligible to get the marks of this range.	If the student has followed the half erroneous approach to attempt the question right he/she is eligible to the marks of this range.	If the student has followed the improper (partly) approach to attempt the question right he/she is eligible to get the marks of this range.	If the student has followed the wrong approach to attempt the question right he/she is eligible to get the marks of this range.

Legibility(2.5 marks)	If the student has attempted the question with complete correctness on legible writing he/she is eligible to get the marks of this range.	If the student has attempted the question with complete correctness on satisfactory writing he/she is eligible to get the marks of this range.	If the student has attempted the question with complete correctness on not satisfactory writing he/she is eligible to get the marks of this range.	If the student has attempted the question with complete correctness on illegible writing he/she is eligible to get the marks of this range.	If the student has attempted the question with complete correctness on poor writing he/she is eligible to get the marks of this range.

CAC-4

Assignment Component: Video Presentation on any one topic from Unit 5 contents

Assignment Description:

To evaluate the student's skills on:

- Capacities to communicate information gained and answer queries behind them.
- Demonstrate the ability to evaluate, integrate, and apply appropriate information from various sources to create cohesive and persuasive arguments.

Evaluation Criteria/Rubric/s:

Video Presentation Rubric	**0 – 2 Marks** **Beginning**	**3 – 5 Marks** **Developing**	**6 – 8 Marks** **Accomplished**	**8 – 10 Marks** **Distinguished**
Content Does the presentation cover require elements thoroughly(2 marks)	Little to no use of scholarly information in the development of the program. Program does not address all	Minimal evidence of use of scholarly information in the development of the program Program addresses all required elements but lacks	Program demonstrates moderate use of scholarly information in the development of the program. Program addresses all required elements. Some elements are covered with	Program demonstrates sound use of scholarly information in the development of the program. All required elements are covered thoroughly.

	the required elements	thoroughness and clarity	excellent clarity and thoroughness while others could be improved.	
Content & Organization Is the topic organized and well thought out(2 marks)	Not organized. Difficult to follow. Poor quality shows poor effort.	Portions may be poorly organized. Hard to follow the progressions of the story. Shows some effort.	Fairly well organized. Format is easy to follow. Good explanation shows good effort.	Program shows a continuous progression of ideas and tells a complete, easily followed story. Well organized. Excellent, well thought out explanation shows superior effort.
Usefulness Does your video stay focus on an informative topic? Does it promote the use of technology to inform the audience about the topic(2 marks)	Topic's usefulness is in question. Does not inform; does not stay focused on the topic.	Topic provides some information and uses technology; has problems staying focused on topic.	Topic is focused and informative; promotes the use of computer technology to create the video to deliver information.	Topic is focused and very informative; promotes the use of computer technology to create the video and makes others want to use the same type of format in delivering information to an audience.
Creativity & Elements of Design Is your video interesting? D	Use of design elements detracts from video and	Minimal use of design elements. Minimal creativity demonstrated.	Good use of graphics and/or other design elements. Good creativity	Excellent sense of design. Excellent demonstration of creativity. Effective camera

oes your video demonstrate creativity in presenting your information? Is the video of good presentation quality(2 marks)	from content. Littl e to no creativity demonstrate d. Sound quality is poor. Video may be out of focus or "shaky".	No transitions. Sou nd is lacking or inappropriate or scratchy. Some pictures or video clips may be out of focus or "shaky".	demonstrated. Some transitions are inappropriately placed. Sound quality is OK. Video is clear and in focus.	techniques used for the video and pictures. Video and focus are of good quality. Smooth transitions are appropriate and aid in the delivery of the presentation.
Overall presentation Presentation meets acceptable time frame? Evidence of participation by all group members? Able to respond to the questions by the students and instructor with ease(2 marks)	Poor topic; Topic falls short of required 4–5-minute time frame. No evidence of participation by all group members. Confusion-unable to respond effectively to questions.	Fair topic. Topic does not meet required 4–5-minute time frame. Weak evidence of participation by all group members. Some difficulty effectively responding to questions.	Good Topic. Falls within 4–5-minute time frame but could have used more or less time to communicate effectively. Evide nce of participation by all all-group members. Respond to questions.	Excellent Topic. Falls within assigned 4–5 min time frame. Evidence of strong participation by all group members. Responded easily to questions.